YITIAN WANG

+1(929)624-1021 cosmotim.github.io
ywang1057@ucr.edu linkedin.com/in/tim-wang-yitian

SUMMARY

Ph.D. scholar in Electrical Engineering specializing in thermal transport and lattice dynamics in functional oxides and solid-state ionic conductors. Experienced in crystal and thin-film growth, vacuum deposition, material characterization, and data-driven modeling using Python and MATLAB. Published 5 first-author works in leading journals and skilled at collaborating with national labs and cross-disciplinary device and materials teams. Additional teaching assistant experience during Ph.D. studies.

EDUCATION

University of California, Riverside, CA Mar. 2021 – Jun. 2026 (expected)
Ph.D. in Electrical and Computer Engineering

Columbia University in the City of New York, NY Aug. 2019 – Feb. 2021
Master of Science, Material Science and Engineering

University of California, Berkeley, CA Jul. 2017 – Aug. 2017
Summer Session, Physics

Beijing Normal University, CN Sep. 2015 – Jun. 2019
Bachelor of Science, Physics

SKILLS

Software	Python, MATLAB, R, SQL, LaTeX, COMSOL, SolidWorks, Linux
Analysis Tools	GSAS-II, Mantid, ImageJ, VESTA, Quantum ESPRESSO, VASP
Synthesis & Growth	Floating-Zone Crystal Growth, PVD / Magnetron Sputtering, CVD
Characterization	SEM, TEM, Scanning Tunneling Microscopy (STM), XRD, AFM, Inelastic Neutron Scattering, Raman, FTIR
Transport Properties	PPMS, DSC, TGA, Four-Probe Transport, Mechanical Strength Testing
Battery & Electrochem	Prototype Cell Assembly, Separator Fabrication, EIS, Galvanostatic Cycling, Thermal Abuse Testing
Methods	Design of Experiments (DOE), AI Agent Training

AWARDS & ACTIVITIES

- Dissertation Completion Fellowship Award, University of California, Riverside (2025)
- MRS 2023 Spring Meeting Highly Commended Student Talk Award
- ‘Jingshi’ Scholarship, Beijing Normal University (2016 – 2018)
- COMAP Mathematical Contest in Modeling — team member and team leader

RESEARCH EXPERIENCE

Graduate Research Assistant Jun. 2021 – Jun. 2025
University of California *Riverside, CA*

· Pioneered lattice-dynamics studies on Li-ion solid-state electrolytes, extending work previously limited to Cu- and Ag-ion systems.
· Published 5 first-author papers in journals including *PRX Energy* and *J. Mater. Chem. A*.
· Built a floating-zone crystal growth workflow from scratch, producing centimeter-scale LLZTO single crystals that enabled phonon-resolved neutron scattering measurements previously infeasible on polycrystalline samples.

- Identified *diffuson*-mediated thermal transport in Li-ion conductors, providing the first verification of this mechanism in Li-ion solid-state electrolytes.
- Developed a two-channel (phonon + *diffuson*) thermal-conductivity model in MATLAB that explained anomalous temperature dependence in Li-ion conductors where the traditional Debye model failed.

Experiment Lead Jan. 2022 – Jul. 2023
Oak Ridge National Laboratory *Oak Ridge, TN*

- Authored two successful beam time proposals and led inelastic neutron-scattering experiments on LLZTO single crystals using TAX (HFIR) and ARCS (SNS) at Oak Ridge National Laboratory (ORNL).
- Resolved the full phonon dispersion of a garnet solid electrolyte for the first time, providing experimental benchmarks for ab initio lattice-dynamics calculations.
- Built automated data-reduction pipelines in Mantid and Python, significantly reducing manual analysis time.
- Managed on-site experimental decisions and coordinated across $\sim$7 research groups spanning UCR, UT Austin, ORNL, ETH Zürich, and TU Wien.

Student Researcher Sep. 2019 – Jan. 2021
Columbia University *New York, NY*

- Engineered composite Li-ion battery separators (γ-C_3N_4/PVDF) via solution processing, targeting improved electrochemical reaction kinetics.
- Assembled and tested CR2032 full cells over 100+ galvanostatic cycles, characterizing cycling performance and interfacial stability via EIS.
- Applied first-principles DFT (Quantum ESPRESSO) to model defect stability in olivine cathode structures, complementing experimental work with computational materials design.

Student Researcher Sep. 2018 – May 2019
Beijing Normal University *Beijing, CN*

- Designed a uniaxial-strain apparatus in SolidWorks exploiting differential thermal contraction for controlled detwinning of iron-based superconductors below 140 K.
- Built a Monte Carlo simulation in MATLAB to model atomic movement of metal atoms inside multi-wall carbon nanotubes, reproducing experimental observations.

TEACHING EXPERIENCE

Graduate Teaching Assistant **University of California, Riverside**

Technology in the Premodern World (ENGR 108)	Reader	Fall 2025	
Circuits and Electronics Lab (EE 005)	Lab Instructor	Spring 2025	*7.0/7.0 (96th %ile)*
Technology in the Premodern World (ENGR 108)	TA / Reader	Spring 2025	*5.7/7.0 (83rd %ile)*
Linear Methods for Engr. Analysis – MATLAB (EE 020B)	Discussion TA	Winter 2023	*6.5/7.0 (81st %ile)*
Magnetic Materials (EE 139)	Discussion TA	Fall 2022	*6.5/7.0 (77th %ile)*

- Served 280+ students across 7 sections over 3 academic years.
- Achieved perfect 7.0/7.0 evaluation (96th percentile campus-wide) in EE 005, with consistently above-department averages across all courses.
- Selected student feedback: “singlehandedly one of the most effective teachers on this campus” and “top tier educator” (EE 005); “the backbone of the entire class” (EE 020B); “always helpful... feedback was helpful in creating a well organized presentation” (EE 139).

INTERNSHIP EXPERIENCE

Administrative Assistant, Physics Department Sep. 2018 – Jan. 2019
Beijing Normal University *Beijing, CN*

- Managed archival of 3 years of departmental paperwork and organized the academic calendar for 4 course sections.

Student Intern Jul. 2016 – Aug. 2016
Shanghai East China Computer Corp. *Shanghai, CN*

· Gained hands-on exposure to IT industry workflows, strengthening teamwork and cross-functional communication skills.

SELECTED PUBLICATIONS

1. **Y. Wang**, Q. Jia, S. Li, L. Shi, Y. Li, X. Chen. "Low thermal conductivity and lattice anharmonicity of NaSICON-type solid electrolyte $Na_3Zr_2Si_2PO_{12}$." *Tungsten*, 2025. [Link]

2. **Y. Wang**, Q. Jia, S. Li, L. Shi, Y. Li, X. Chen. "Glass-like thermal transport in polycrystalline perovskite Li-ion conductor $Li_{3/8}Sr_{7/16}Hf_{1/4}Ta_{3/4}O_3$." *Chem. Commun.*, 2025. [Link] [Front Cover]

3. **Y. Wang**, Y. Su, J. Carrete, H. Zhang, N. Wu, *et al.* "Origin of intrinsically low thermal conductivity in a garnet-type solid electrolyte." *PRX Energy*, 2025. [Link]

4. S. Li, S. Guo, **Y. Wang**, H. Li, Y. Xu, V. Carta, J. Zhou, X. Chen. "Enhanced magnon thermal transport in yttrium-doped spin ladder compounds $Sr_{14-x}Y_xCu_{24}O_{41}$." *J. Appl. Phys.*, 2024. [Link]

5. **Y. Wang**, S. Li, N. Wu, Q. Jia, T. Hoke, L. Shi, Y. Li, X. Chen. "Thermal properties and lattice anharmonicity of Li-ion conducting garnet solid electrolyte $Li_{6.5}La_3Zr_{1.5}Ta_{0.5}O_{12}$." *J. Mater. Chem. A*, 2024. [Link]

6. S. Guo, H. Li, X. Bai, **Y. Wang**, S. Li, R. E. Dunin-Borkowski, J. Zhou, X. Chen. "Size-dependent magnon thermal transport in a nanostructured quantum magnet." *Cell Rep. Phys. Sci.*, 2024. [Link]

7. Y. Xu, Z. Barani, P. Xiao, S. Sudhindra, **Y. Wang**, *et al.* "Crystal structure and thermoelectric properties of layered van der Waals semimetal $ZrTiSe_4$." *Chem. Mater.*, 2022. [Link]

8. **Y. Wang**, X. Chen. "Single crystal growth and electrochemical studies of garnet-type fast Li-ion conductors." *Tungsten*, 2022. [Link]